

Figure 9: Set the URL to the Firebase component

- Upon clicking the *Send chat* button, we will first validate whether anything is entered in the chat box.
- First, the procedure will check if there is anything entered in the chat box.
- On the next line, it will call the next procedure to check for the name of the user.
- If everything seems fine, it will store the chat message appended with the user name in the Firebase database. You are already aware of the tag and value terms of this database.
- The greatest feature of the Firebase database is its concurrency and real-time updates.
- Now you are done with the block editor too. Next, let's move to download and install the app on your phone to check how it works.

Packaging and testing

To test the app, you need to get it on your phone. First, download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the 'Build' button. It will show you the option to download the *apk* to your computer.
2. You will be able to see the progress of the downloading process and after that's successfully done, the application will be placed in the



Figure 10: Block editor Image 1

Download folder or the preferred location you have set for it.

3. Now you need to get this *apk* file to your mobile phone either via Bluetooth or via USB cable. Once you have placed the *apk* file on your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying that the install is from an untrusted source. Allow this from the settings and after successful installation, you will see the icon of your application in the menu of your mobile. Here, you will see the default icon, which can be changed and we will tell you how to do this as we move ahead in this course.

I hope your application is working exactly as per the requirements you have



Figure 11: Block editor Image 2

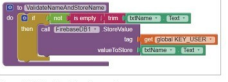


Figure 12: Block editor image 3



Figure 13: Block editor image 4

set. Now, you can customise various things like the image, sound and behaviour also.

Debugging the application

We have just created the prototype of the application with very basic functionality and what the user may be interested in. Now come the various use cases that require attention, so that our app does not annoy the user but is able to sustain itself. Consider the following scenarios:

1. What would happen if two users are clicking on the *Send* button at exactly the same time? Do you think the current implementation is feasible enough to handle the confusion?
2. Can we create multiple groups and store the chats into different project buckets?
3. Can we think of sharing media as well?
4. What if the device of the user is not connected to the Internet? Will it be able to sync all the previous chats upon connecting to the Internet?

These are some of the scenarios that might occur and the user will be pretty happy seeing them addressed. Think about how you can integrate these into the application. Do ask me if you fail to accomplish any of the above cases.

You have successfully built another useful Android app for yourself. Happy inventing!

Feedback on your interests and ideas on how to make Android applications will be much appreciated. 

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